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### Label, peeling, universal printheads and related methods

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#### Abstract

An example disclosed printer configured to peel media from a backing, the printer having a peeler assembly that is engageable between a peeling position, wherein the printer is configured to peel the media from the backing, and a non-peeling position, wherein the printer is not configured to peel the media from the backing. The printer further including a sensor configured to send a signal corresponding to a position of the peeler assembly. The printer further including a communication interface configured to receive the position of the peeler assembly from the sensor and configured to transmit the position of the peeler assembly to a secondary component.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This patent arises from a continuation of U.S. patent application Ser. No. 17/229,675, filed on Apr. 13, 2021, which is a continuation of U.S. patent application Ser. No. 15/957,129, filed on Apr. 19, 2018, which is a continuation of U.S. patent application Ser. No. 15/211,615, filed on Jul. 15, 2016, now U.S. Pat. No. 9,975,360, which is a continuation of U.S. patent application Ser. No. 13/725,247, filed on Dec. 21, 2012, now U.S. Pat. No. 9,434,191, which is a continuation-in-part of U.S. patent application Ser. No. 13/085,422 filed on Apr. 12, 2011, now U.S. Pat. No. 8,714,851, which claims the benefit of priority to U.S.

## FIELD

(1) Embodiments discussed herein are related to printers and, more particularly, to systems, methods, apparatuses, computer readable media products and other means for providing printheads and assemblies that are configured to peel media units from backing.

## BACKGROUND

(2) Printers are designed and known to be used in traditional office environments. Some printers have a more portable design that allow them to be used for many other applications beyond traditional office printing, such as printing customer receipts at the point of delivery, price tags at product display shelves, shipment labels, parking garage receipts, adhesive labels, law enforcement tickets, and gas and utility inspection documents at people's homes. Through applied effort, ingenuity, and innovation, various printer improvements are embodied by the present invention, examples of which are discussed below.

## SUMMARY

(3) Some embodiments discussed herein are related to a printer assembly configured to engage a label and peel the label from a backing without the backing having to be threaded through the printer assembly. The printer assembly may comprise one or more peel bars, support members, actuating shafts, and gears, among other things.

(4) An example embodiment of the present invention may include a printer configured to print to a media disposed on a backing. The printer including a housing including a media cover, a media cover release actuator, and a peeler assembly actuator. The media cover release actuator and the peeler assembly actuator may each be accessible for actuation by a user when the media cover is in a closed position. The media cover release actuator may be configured to release the media cover from the closed position in response to actuation by a user. The peeler assembly actuator may define an engaged position and a disengaged position. The printer may include a peeler assembly at least partially enclosed by the housing and engagable between a peeling position, where the printer is configured to peel the media from the backing, and a non-peeling position, where the printer is not configured to peel the media from the backing. The peeler assembly actuator may be configured to, in response to actuation by the user from the disengaged position to the engaged position, transition the peeler assembly from the non-peeling position to the peeling position. The peeler assembly actuator may be configured to preclude actuation of the media cover release actuator when the peeler assembly actuator is in the engaged position.

(5) Embodiments may further include a lock mechanism configured to lock the peeler assembly actuator in the engaged position. The lock mechanism may include a lock button. The peeler assembly actuator may be permitted to move from the engaged position to the disengaged position in response to the locking button being depressed. The media cover may be released in response to the media cover release actuator being actuated through a range of motion, where the peeler assembly actuator may be configured to block the media cover release actuator from being actuated through the range of motion when the peeler assembly actuator is in the engaged position.

(6) According to some embodiments, the peeler assembly may include a peel bar, a support member connected to the peel bar, and an actuating shaft engaged with the support member. The peeler assembly actuator may be configured to drive the rotation of the actuation shaft in response to the peeler assembly actuator being moved from the disengaged position to the engaged position. The support member may be attached to a printer chassis at a pivot point and the actuation shaft may be configured to pivot the support member about a pivot point in response to being rotated. The media cover may define an open position and the peeler assembly in the peeling position may preclude the media cover from being moved from the open position to the closed position.

(7) An example embodiment of the invention may include a peeler assembly configured to peel print media from a backing, for use in a printer that includes a media cover and a media cover release actuator. The peeler assembly may include a peeler assembly actuator defining an engaged position corresponding to a peel position of the peeler assembly and a disengaged position corresponding to a non-peeling position of the peeler assembly, where the peeler assembly actuator may be configured to preclude actuation of the media cover release actuator when the peeler assembly actuator is in the engaged position. A lock mechanism may be configured to lock the peeler assembly actuator in the first position.

(8) Embodiments of the peeler assembly may include a first support member, a second support member, and a peel bar that extends proximate the first support member and the second support member, substantially perpendicular to the first support member and the second support member. Embodiments may further include an actuating shaft, where the actuating shaft extends between the first support member and the second support member. The peel bar may be disposed between a first end of each of the first support member and the second support member and the actuating shaft may be coupled to the first support member and the second support member proximate a second end of each of the first support member and the second support member. The first support member and the second support member may each be configured to be coupled to a printer chassis at a pivot point between each of their respective first end and second end. In response to rotation of the actuating shaft, each of the first support member and the second support member may be rotated about their respective pivot points. In response to the first and second support members rotating about their respective pivot points, the peel bar may be moved between the peeling position and the non-peeling position.

(9) Embodiments of the present invention may provide a printer that includes a housing, a media cover, a peel bar, a peeler assembly actuator, and a media cover release actuator. The media cover may be configured to be moved between an open position and a closed position. The peeler assembly may be configured to be moved between a peeling position and a non-peeling position. The peeler assembly actuator may be configured to move the peeler assembly between the peeling position and the non-peeling position, where the peeler assembly actuator defines an engaged position corresponding to the peeling position of the peeler assembly and a disengaged position corresponding to the non-peeling position of the peeler assembly. The media cover release actuator may be configured to release the media cover from the closed position in response to the media cover release actuator being actuated, where the media cover release actuator is precluded from being actuated when the peeler assembly actuator is in the first position.

(10) Embodiments may further include a locking mechanism defining a locked position and an unlocked position, where the peeler assembly actuator is precluded from moving from the engaged position to the disengaged position in response to the locking mechanism being in the locked position. The media cover may be precluded from moving between the open position and the closed position in response to the peeler assembly being in the peeling position. The peeler assembly actuator may move between the engaged position and the disengaged position along a first axis, and the media cover release actuator may be configured to be actuated along a second axis, substantially parallel to the first axis.

(11) According to some embodiments, the media cover release actuator and the peeler actuator may each be accessible from outside the housing when the media cover is in the closed position. The peeler assembly actuator may be configured to engage a locking mechanism in response to the peeler assembly actuator being moved from the disengaged position to the engaged position. The peeler assembly actuator may be configured to be movable from the engaged position to the disengaged position in response to the locking mechanism being moved from a locked position to an unlocked position. The locking mechanism may be moved from the locked position to the unlocked position in response to a user actuating a lock button of the locking mechanism.

(12) Embodiments of the present invention may provide a printer including a housing, a media

cover configured to be moved between an open position and a closed position, and a peeler assembly configured to move between a peeling position and a non-peeling position. The peeler assembly may be configured to be movable from the non-peeling position to the peeling position when the media cover is in the closed position. The printer may further include a peeler assembly actuator configured to move the peeler assembly between the peeling position and the non-peeling position. The peeler assembly actuator may be configured to, in response to actuation by the user from a disengaged position to an engaged position, transition the peeler assembly from the non-peeling position to the peeling position. The peeler assembly actuator may be configured to be accessible when the media cover is in the closed position.

(13) According to some embodiments, the printer may include a locking mechanism defining a locked position and an unlocked position, where the peeler assembly actuator may be precluded from moving from the engaged position to the disengaged position in response to the locking mechanism being disposed in the locked position. The printer may be configured to print to a continuous strip of media, where the media is threaded around the peel bar in response to the peeler assembly moving from the non-peeling position to the peeling position.

(14) Embodiments of the present invention may provide a printer that includes a peeler assembly that is engagable between a peeling position, where the printer is configured to peel the media from the backing, and a non-peeling position, where the printer is not configured to peel the media from the backing. The printer may further include a sensor configured send a signal corresponding to a position of the peeler assembly, and a printer controller configured to receive the position of the peeler assembly from the sensor and configured to adjust at least one print setting in response to receiving the signal corresponding to the position of the peeler assembly. The at least one print setting may include the print speed, the printhead temperature, or the printhead position. The sensor may include a binary switch operable to detect the peeling position of the peeler assembly or the non-peeling position of the peeler assembly. The sensor may include a proximity sensor.

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## Description

### BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING(S)

(1) Having thus described the invention in general terms, reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:

(2) FIG. 1 shows an isometric view of a mobile printer in accordance with additional embodiments discussed herein;

(3) FIG. 2 shows an isometric view of a mobile printer in accordance with some embodiments discussed herein;

(4) FIG. 3 shows a front view of the mobile printer discussed in connection with FIG. 2;

(5) FIG. 4 shows a back view of the mobile printer discussed in connection with FIG. 2;

(6) FIG. 5 shows a left side view of the mobile printer discussed in connection with FIG. 2;

(7) FIGS. 6A and 6B each show a right side view of a mobile printer in accordance with example embodiments of the present invention;

(8) FIG. 7 shows a top view of the mobile printer discussed in connection with FIG. 2;

(9) FIGS. 8A-8D show a series of isometric views of the mobile printer discussed in connection with FIG. 2 to illustrate a process of loading print media into the mobile printer and engaging the media using a peeler in accordance with some embodiments discussed herein;

(10) FIGS. 9A and 9B show an example peeler assembly in accordance with some embodiments discussed herein;

(11) FIGS. 10A-10C show an example process for transition of peeler assembly of FIGS. 9A and 9B from a non-peeling position to a ready position (or vice-versa) and from the ready position to a peeling position (or vice-versa);

- (12) FIGS. **10D** and **10E** show another example peeler assembly in accordance with some embodiments discussed herein;
- (13) FIGS. **10F-10H** show an example process for transitioning the peeler assembly of FIGS. **10D** and **10E** from a non-peeling position to a ready position (or vice-versa) and from the ready position to a peeling position (or vice-versa);
- (14) FIGS. **10I-10L** show another example the peeler assembly and process for transitioning from a non-peeling position to a ready position (or vice-versa) and from the ready position to a peeling position (or vice-versa);
- (15) FIGS. **10M-10N** illustrates another example of a peeler assembly and process for transitioning from a peeling position to a non-peeling position (or vice-versa);
- (16) FIGS. **10P-10R** illustrate an example embodiment of a printer implementing the peeler assembly of FIGS. **10M-10N**;
- (17) FIG. **10S** illustrates a cross section of a portion of a printer according to an example embodiment of the present invention depicting a media path for a peeler assembly in a non-peeling position;
- (18) FIG. **10T** illustrates a cross section of a portion of a printer according to an example embodiment of the present invention depicting a media feed path for a peeler assembly in a peeling position;
- (19) FIG. **11A** shows an isometric view of a printer in accordance with some embodiments discussed herein;
- (20) FIG. **11B** shows an example block diagram of circuitry that may be included in some embodiments discussed herein;
- (21) FIG. **12** shows a process flow related to using a peeler bar that may be executed by a processor and/or other hardware at least partially implemented in a printer in accordance with some embodiments discussed herein;
- (22) FIG. **13** shows a process flow related to using a peeler sensor that may be executed by a processor and/or other hardware at least partially implemented in a printer in accordance with some embodiments discussed herein; and
- (23) FIGS. **14A-14C** show examples of a universal printhead that may be incorporated in a printer in accordance with some embodiments discussed herein.

#### DETAILED DESCRIPTION

(24) The present invention now will be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all embodiments of the inventions are shown. Indeed, these inventions may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will satisfy applicable legal requirements. Like numbers refer to like elements throughout.

#### Overview of Example Printer Platforms

(25) FIG. **1** shows mobile printer **100**, which is in accordance with some embodiments of the present invention. Mobile printer **100**, like some of the other exemplary printers discussed herein, can weigh less than 1.6 pounds, can have a volume of about 61 cubic inches, and can be durable enough to function properly after falling from a height of at least 5 feet. Mobile printer **100** can also be used to print user-readable indicia at, e.g., a speed of one or more inches per second. In some embodiments, the torque and/or print speed can be dynamically adjusted based upon drive motor and/or any other type of the printing parameters selected in response to the printer's processor determining, e.g., the type of media (e.g., backless media, media with backing to be peeled, among others). Mobile printer **100** can also be configured to encode machine-readable indicia onto media.

(26) The relatively small profile and increased durability of mobile printer **100** can be enabled by, among other things, the arrangement of the internal circuitry and/or circuit boards onto which the

circuitry is mounted. For example, the circuitry of mobile printer **100** can be laid out on a plurality of circuit boards (instead of a single circuit board). Additional examples of how circuitry may be arranged on one or more circuit boards are discussed in commonly-assigned U.S. patent application Ser. No. 13/085,438, titled “PRINTER MOBILITY AND SCALABILITY,” which is incorporated herein in its entirety by reference.

(27) The media can include, for example, a number of adhesive-backed labels supported by a backing or other carrier. The backing is typically a paper strip coated with silicone to facilitate easy removal of the adhesive backed label, commonly called liner, but could be any other type of carrier, even a multi-layer or “piggyback” labels designed for specific printing of multi-part documents. In RFID applications, the labels may include an RFID transponder or other type of circuitry (sometimes referred to herein as an “inlay”). A peeler (discussed further in connection with FIGS. **7-13**) and/or other component(s) can be included in mobile printer **100** and may be used to separate the media from the backing after printing/encoding. The media can include a single media unit, or the media can include individual media units that are rolled together, fan-folded, or otherwise assembled together, and inserted into mobile printer **100**. For example, media cover **102** can open (similar or the same as that shown in FIGS. **8A** and **8B**) allowing mobile printer **100** to receive a single media unit, a roll of media units, a fanfold of media units, or any other suitable arrangement of one or more media units. Mobile printer **100** can then feed the media through media slot **104**. In some embodiments, a tear bar or other type of cutting component (discussed below) can be incorporated near media slot **104** to help the user remove a label, receipt, or other type of media from the roll after printing/encoding.

(28) A peeler assembly, such as that discussed in connection with FIGS. **9A-10C** and/or **10D-10T**, may also be included in mobile printer **100**. The peeler assembly may comprise a peel bar and/or other component(s) (including those discussed in reference to FIGS. **8A-10R**), and be positioned proximate media slot **104**. Mobile printer **100** may include gears and/or other components that are adapted to engage the peeler to print media (such as, e.g., media cover **102** may be configured to latch the peel bar in a peeling position), subsequent to the peel bar being released from its stowed or other type of non-peeling position by a user and/or mobile printer **100**. The peeler assembly can then be used to at least partially remove a label or other type of media from any type of backing, such as a media liner, after printing/encoding the media.

(29) Housing **106**, including media cover **102**, can be made from any suitable material and/or combinations of materials. For example, housing **106** can be made from plastic(s), rubber, metal, composite material, any other type of material, or combination thereof (such as, e.g., a rubber-infused plastic). Housing material **106** can be strong enough to protect the internal components from a fall, while still allowing wireless signals to radiate through in at least some locations. Housing **106** is shown in FIG. **1** as having a center seam between top half **130** of the housing and bottom half **132** of the housing. Other embodiments, examples of which are discussed below (in connection with, e.g., FIGS. **4-8B**), include a seam line that allows, among other things, the bottom portion of the housing to be the same size and shape, regardless of the overall shape and size of printers. Additional examples of defining shells and other shape defining housing portions of printers that may be used in some embodiments discussed herein are provided in commonly-assigned U.S. patent application Ser. No. 13/085,438, titled “PRINTER MOBILITY AND SCALABILITY,” which is incorporated herein in its entirety by reference.

(30) Mobile printer **100** can also include one or more user input/output components, such as display **108**, left button **110**, right button **112**, menu button **114**, back button **116**, navigation buttons **118**, feed button **120**, power button **122**, and cover release actuator **124**. Other examples of buttons, sensors and other type of user and environmental inputs may be included, some of which are discussed herein. For example, additional user inputs and/or outputs may be available to the user only when media cover **102** is open. Additional examples of printer user interfaces that may be used in some embodiments discussed herein are provided in commonly-assigned U.S. patent

application Ser. No. 13/085,438, titled “PRINTER MOBILITY AND SCALABILITY,” which is incorporated herein in its entirety by reference.

(31) Display **108** can include components configured to receive data inputs and present a display. In some embodiments, display **108** may also include components that enable touch-sensitive functionality. The touch-sensitive components of display **108** can include, for example, capacitance sensor(s), resistance sensor(s), acoustic wave sensor(s), optical sensor(s), any other type of sensor(s), and/or combination thereof. Display **108** can also include a liquid crystal display (“LCD”), light emitting diode (“LED”) display, an organic light emitting diode (“OLED”) display, any other type of display, or combination thereof. Different types of displays have various advantages and disadvantages relative to each other. For example, a 2.1 inch OLED display may have a relatively larger viewing angle than a 2.1 inch LCD. However, as known to those skilled in the art, OLED displays are currently more expensive than comparable LCDs. Other examples of displays that may be used by some embodiments discussed herein include an electronic paper display (“EPD,” sometimes referred to as electronic ink or e-ink) or other type of bi-stable display(s). Power consumption, brightness, readability (in sunlight, darkness, etc.), expected life span, and other factors may also differ among different types of displays and impact the best display to be used as display **108** depending on the intended usage of mobile printer **100**. In some embodiments, an ambient light sensor and/or other type of sensor can be integrated into the display and/or other portion of printer **100**, such as in and/or near the peeler bar or media cover. The one or more sensors can be used to, for example, adjust the brightness of the display, detect a label that needs to be removed from the peel bar, and determine when printer **100** is moving, among other things.

(32) Display **108** can be configured to present an icon based menu and/or any other type of menu hierarchy. Among other things, display **108** can be adapted to display Asian-language fonts in relatively high resolution. Various fonts, firmware and/or other data may be downloaded onto printer **100** (some examples of which are discussed further below).

(33) In some embodiments, display **108** may only function as an output display component and be unable to function as an input component. For example, display **108** may lack a functioning touch-sensitive input component and/or the appropriate software/hardware/firmware necessary to enable the touch-sensitive input component.

(34) Regardless of whether display **108** includes a touch sensitive component that displays user-selectable buttons, mobile printer **100** can include “soft” keys, such as left button **110** and right button **112**, which are hardware-based keys (as opposed to software-based keys presented by a touch-sensitive display) that can be used to select options presented by display **108**. In some embodiments, portions of the screen of display **108** can be dedicated to and associated with left button **110** and/or right button **112**. For example, the area of display **108** located immediately above left button **110** can be used to present an option that can be selected in response to left button **110** being depressed. Likewise, the area of display **108** located immediately above right button **112** can be used to present another option that can be selected in response to right button **112** being depressed. The options presented in the screen areas associated with left button **110** and/or right button **112** can be dynamic and change based upon the current display, allowing left button **110** and right button **112** to provide flexible navigation of the menu hierarchy. Additional examples of printer user interfaces that may be used in some embodiments discussed herein are provided in commonly-assigned U.S. patent application Ser. No. 13/085,438, titled “PRINTER MOBILITY AND SCALABILITY,” which was incorporated herein in its entirety by reference.

(35) Mobile printer **100** can also include menu button **114**. Internal circuitry (e.g., the processor and/or other components, such as those discussed in connection with FIG. **11B**) of mobile printer **100** can be configured to, for example, present a main menu or other type of display on display **108** in response to menu button **114** being depressed. Software and/or firmware, which include coded machine-readable instructions for executing the functionality of mobile printer **100**, can be stored



in memory or any other type of computer readable media device included in mobile printer **100**. For example, mobile printer **100** may include 128 megabytes of memory, 256 megabytes of memory, or any other suitable amount of removable or embedded memory in the form of volatile and/or non-volatile storage (e.g., flash memory, magnetic disk memory, etc.).

(36) Back button **116** can cause the internal circuitry to present a display that is higher up a menu hierarchy of mobile printer **100**. In other embodiments, back button **116** (or another button which is not shown) can be used to physically “back-up” or otherwise control the movement of the media being printed/encoded by mobile printer **100**. In yet other embodiments, back button **116** (and/or any other component of mobile printer **100**) can have dynamic functionality, in that selecting back button **116** causes mobile printer **100** to respond differently depending on a range of variables. For example, back button **116** may cause display **108** to advance up the menu hierarchy while media cover **102** is in a closed position (as shown in FIG. 1) and cause the printing media to back-feed while media cover **102** is in an open position (as shown in FIGS. 8A and 8B), or vice-versa.

(37) Navigation buttons **118** are shown in FIG. 1 as comprising four directional buttons and a center button. Navigational buttons **118** can enable a user to, for example, move a cursor among and/or select one or more options presented by display **108**. Navigational buttons **118** can also provide an intuitive interface for allowing a user to move to and/or select an option using fewer key presses. Additional examples of printer user interfaces that may be used in some embodiments discussed herein are provided in commonly-assigned U.S. patent application Ser. No. 13/085,438, titled “PRINTER MOBILITY AND SCALABILITY,” which was incorporated herein in its entirety by reference.

(38) Mobile printer **100** can be powered ON and OFF using power button **122**. In some embodiments, power button **122** can be used to cause mobile printer **100** to enter or exit a standby mode. For example, in response to power button **122** being depressed for less than a predetermined period of time (e.g., 5 seconds), mobile printer **100** can either enter or exit standby mode (depending on whether or not mobile printer **100** is currently in an active or standby mode). But in response to power button **122** being depressed for more than 5 seconds, mobile printer **100** can power OFF (if ON). In some embodiments, the circuitry of mobile printer **100** can be configured to automatically power OFF after a predetermined period of time or in response to determining the battery power has dropped below a predetermined threshold. Mobile printer **100** may also be configured to power ON automatically, for example, in response to a print command being issued by another device. In some embodiments, mobile printer **100** can be configured to automatically enter and/or exit a stand-by or other power-saving mode (including dimming the display screen, turning OFF wireless components, and/or execute other power reduction configuration settings). For example, a power-saving mode may be entered or exited after a predetermined period of time has elapsed and/or an environmental trigger has been detected (e.g., light detected by an ambient light sensor, movement detected by a jiggle switch, accelerometer and/or other type of movement sensor, etc.).

(39) Cover release actuator **124** can be used to unlock and/or open media cover **102**. When media cover **102** is open (as shown in FIGS. 8A and 8B), media can be loaded into mobile printer **100**, media jams can be fixed, ribbon or other printing components can be replaced, the peeler bar can be released to engage the media, etc.

(40) The relatively small size of mobile printer **100** allows mobile printer **100** to be attached, mounted, or otherwise physically coupled to a number of devices. For example, mobile printer **100** can be attached to a fork lift (or other warehouse apparatus), automobile (e.g., police car), healthcare device, shopping cart, belt loop, belt, and lanyard, among other things. To facilitate its mechanical or other type of physical coupling to another apparatus, mobile printer **100** can include channels **126**, which are adapted to receive a mounting component (e.g., a shoulder strap, belt, or other type of tether), locking component, and/or other type of apparatus(es). Other mounting components, including those used for a dock, are provided in commonly-assigned U.S. patent

application Ser. No. 13/085,431, titled "MOBILE PRINTER NETWORKING AND INTERFACING," which was incorporated herein in its entirety by reference.

(41) Protective cover **128** can be used to prevent water, dirt and other elements from entering one or more electrical coupling components of mobile printer **100**. For example, one or more universal serial bus ("USB") ports, mini-USB ports, a serial enhanced security ports, an Ethernet ports, an optical ports, and/or any other type of input components, output components and/or input/output components may be located behind protective cover **128**. Media cover **128** can be removed and/or opened to access the protected component(s). The protected component(s) can also incorporate, for example, strain relief technology, some examples of which are provided in commonly-assigned U.S. patent application Ser. No. 13/085,438, titled "PRINTER MOBILITY AND SCALABILITY," which was incorporated herein in its entirety by reference.

(42) FIG. 2 shows printer **200**, which includes components similar to or the same as those discussed above in connection with mobile printer **100**. To avoid overcomplicating the discussion, like reference numbers refer to like elements throughout the drawings. (Although a display screen is not shown in FIG. 2, reference number 108 points to a portion of housing **106** that may be adapted to receive any type of display or other type of user interface.)

(43) In some embodiments, the portion of printer **200**'s housing where display **108** may be integrated can be formed from one or more different materials than other portions of the housing. For example, the side walls of the housing (such as the portions where channels **126** are located) can be formed from an injection molded plastic, and the portion for display **108** can comprise die cut rubber. When manufacturing printer **200**, for example, plastic for the housing can be injected molded around the die cut rubber, which may allow the same injection mold to be used for printers that have different sized and/or types of input components (e.g., different types of display screens, navigation button arrangements, etc.). In some embodiments, a rubber and/or other type of overmold can be applied to one or more of the printer's side walls and/or other components. Allowing the same molds to be used for multiple products can reduce manufacturing and machining costs sometimes associated with providing various product options to customers. In other embodiments, rather than place display **108** into rubber, the portion of the printer's housing that receives display **108** can be plastic and/or any other type of material. Other design aspects, some of which are discussed in commonly-assigned U.S. patent application Ser. No. 13/085,438, titled "PRINTER MOBILITY AND SCALABILITY," which was incorporated herein in its entirety by reference, may be included in some of the embodiments discussed herein and allow printer components to be used across various types of mobile printers, desktop printers and other devices in accordance with some embodiments of the present invention.

(44) Printer **200** includes clip **202**, which may enable printer **200** to be attached to a user's belt or belt loop. In some embodiments, clip **202** may swivel on a ball hinge or may remain in a fixed position relative printer **200**. Housing **106** of printer **200** can be molded or otherwise adapted to receive clip **202** with or without another component. Some embodiments of housing **106**, such as that shown in FIG. 2 that lacks a seam line running through the ball joint receptacle, can directly receive clip **202** without sacrificing much, if any, strength of the connection, even absent another component. Clip **202** can be removable and/or replaced with one or more other types of attaching components. For example, a magnetic attaching component could be located where clip **202** is shown in FIG. 2, and used to attach printer **200** to a metallic surface. Additional examples of belt clips that may be used in combination with some embodiments discussed herein are provided in commonly-assigned U.S. patent application Ser. No. 13/085,438, titled "PRINTER MOBILITY AND SCALABILITY," which was incorporated herein in its entirety by reference.

(45) FIGS. 3-8D show different views of printer **200**. For example, FIG. 3 shows a front view of printer **200**.

(46) FIG. 4 shows a back view of printer **200**. Hinge **402** can include a spring or other mechanism that allows media cover **102** to be driven open in response to a latch being released. For example,

cover release actuator **124** can be adapted to release such a latch when depressed by a user. When media cover **102** is open, media can be loaded into printer **200**, media jams can be corrected, print ribbon or other consumable printing components can be replaced, among other things. Cover release actuator **124** can also be configured to release a peel bar from a non-peeling position. Cover release actuator **124** can also be configured to release and/or drive a peel bar from a non-peeling position to a ready and/or peeling position. Additional examples of peel bars' functionality, including a examples of the non-peeling, ready and peeling positions, are discussed in connection with, e.g., FIGS. **9A-10R**.

(47) In other embodiments, the peel bar can function independent from cover release actuator **124**. For example, cover release actuator **124** can be configured to open media cover **102** without causing the peel bar to be released from the latched position.

(48) In some embodiments, cover release actuator **124** may be configured to function differently depending on how it is utilized (e.g., depressed). For example, when cover release actuator **124** is depressed partially, media cover **102** may be opened, and when cover release actuator **124** is depressed further (e.g., all or most of the way down), the peel bar is also released to a ready and/or peeling position.

(49) The back view of printer **200** also shows that its housing was assembled from two pieces, namely, defining portion **404** and base portion **406**. Defining portion **404** meets base portion **406** underneath printer **200** (as opposed to along the lateral sides such as housing **106** of mobile printer **100**). Additional examples of printer housings and advantageous realized therefrom are provided in commonly-assigned U.S. patent application Ser. No. 13/085,431, titled "MOBILE PRINTER NETWORKING AND INTERFACING," which was incorporated herein in its entirety by reference.

(50) Hinge **402** can attach media cover **102** to defining portion **404**, and fasteners **408** (which may be screws and/or any other type of mechanical fasteners) can attach defining portion **404** to base portion **406**. In some embodiments, the circuit board(s) of mobile printer **100** can be removed from (e.g., slid out of) housing **106** after base portion **406** is separated from defining portion **404**. Each portion or sub-portion(s) of printer **200**'s housing can be constructed from any type of material, and may have a varying degree of transparency or opaqueness. For example, media cover **102** can be transparent, while the rest of the housing can be opaque.

(51) FIG. **5** shows a left side view of printer **200**, which includes protective cover **502** and alignment cavity **504**. Similar to or the same as protective cover **128**, protective cover **502** may be removed to expose one or more input, output, and/or input/output components that enable a power source, peripheral device, accessory device, network device, and/or other apparatus to be coupled with the circuitry of printer **200**. In some embodiments, printer **200** can be configured to accept and/or work with accessories common to other types or models of devices. For example, a battery charger may be coupled to a port behind protective cover **502** and used to charge the battery pack of printer **200**.

(52) Alignment cavity **504** can be used to facilitate the proper electro-mechanical coupling of printer **200** with one or more accessory devices. For example, a docking station (sometimes referred to more generally herein as a "dock"), charging station, or mobile palette may define a protrusion that corresponds with and fits into cavity **504**. As referred to herein, a "charging station," refers to an apparatus that can function as a source of power for charging the batteries of the mobile printer without facilitating data communications between the mobile printer and a network device. A "docking station," as used herein, refers to an apparatus that can receive and electrically couple with a printer, function as a source of power to charge the printer's batteries, and facilitate data communications between the printer and a host device (e.g., provide Ethernet communications to a network server). A docking station may be associated with a fixed physical location that is known to the host device and can be used to determine the location of printer **200**. Exemplary docking stations are provided in commonly-assigned U.S. patent application Ser. No. 13/085,431, titled

“MOBILE PRINTER NETWORKING AND INTERFACING,” which was incorporated herein in its entirety by reference.

(53) FIGS. 6A and 6B show two right side views of printer **200**, which include cavity **602**. Cavity **602** may be a detent adapted to receive a coupling mechanism, and may be the same as or similar to cavity **504** in design, functionality and/or application. In other embodiments (not shown), cavity **602** can take a different shape and/or form to cause printer **200** to be aligned in a particular manner relative to, e.g., a docking station or other apparatus. In some embodiments, cavities **504**, **602** may operate to mechanically align printer **200** in a desirable presentation manner when printer **200** is docked onto a docking station or the like. By being properly aligned and coupled with a docking station, for example, the location of printer **200** may be determined based on the docking station's known location. In some embodiments, such as those in accordance with FIG. 6B and are provided in commonly-assigned U.S. patent application Ser. No. 13/085,438, titled “PRINTER MOBILITY AND SCALABILITY,” which was incorporated herein in its entirety by reference, one or more notches may also be included in base portion **604**.

(54) Peeler Assembly

(55) Embodiments of the present invention may include a mechanism by which media disposed on a backing is separated from the backing, or peeled, as the backing is advanced after printing. The peeling of media from a backing may be achieved by directing the backing along a tortured path including at least one relatively sharp bend. As the backing passes around this bend, the media, which is releasably adhered to the backing, may not follow the sharp bend of the backing, but instead be separated from the backing in a peeling action. Example embodiments are described herein which include a peeling assembly which is engaged (i.e., moved to the peeling position) in response to a media cover being closed, and an embodiment in which the peeling assembly may only be engaged when the media cover is in the closed position. While embodiments described herein are capable of peeling media from a backing, each embodiment may also be used with the peeler assembly in a non-peeling position, in which a continuous media strip may be printed, and the media may be torn from the strip of media as necessary.

(56) FIG. 7 shows a top view of printer **200**, which includes tear bar **702** that is visible through media slot **104**. Tear bar **702** can be used to tear off or otherwise remove media from a roll or, more generally, from printer **200** after printing/encoding. Peeler assembly **704**, shown in the disengaged, non-peeling position in FIG. 7, can also be incorporated in printer **200**. Peeler assembly **704** may be a component of a threadless peeler assembly in accordance with some embodiments discussed herein.

(57) As noted above, print media, such as adhesive labels or tags, may be disposed on a carrier substrate also known as a backing. Such media may be printed by a portable printer on an as-needed basis such that upon printing of the label or tag, the label or tag is then adhered to a surface. Embodiments may include printers configured to print shipping labels, identification tags, product information, etc. As such, when printing such media, particularly when printing them on an as-needed basis, it may be desirable to have the media separated from the backing automatically rather than requiring a user to individually peel each label or tag from the backing after printing. Thus, example embodiments provided herein may automatically separate the media from the backing during the printing operation. Embodiments may include a peeler assembly, as detailed below, to separate the media from the backing. As the media disposed on the backing is advanced past the printing mechanism, the peel bar of peeler assemblies may engage the media and separate the media from the backing. Although embodiments here are shown with a portable printer, similar peeling mechanisms and methods may be used with printer applicators, industrial printers, automatic label applicators, and similar devices which may not be portable.

(58) FIG. 8A shows an isometric, rear perspective view of printer **200** having media cover **102** in an open position and peeler assembly **704** in an “up” position. Peeler assembly **704** can be stowed in a “down,” non-peeling position and subsequently unlatched into the up position in response to an

intuitive, peeler assembly actuator **802** being manually depressed by a user. When the user is loading linerless media (namely media that is not peeled from a liner or backing after printing, such as a sheet or roll of paper) into printer **200**, peeler assembly **704** can remain latched in a stowed, non-peeling position (as opposed to, e.g., being automatically released in response to media cover **102** being opened). As a result, some embodiments of peeler assembly **704** can realize a number of advantages when implemented in a mobile printer. For example, peeler assembly **704** can be relatively smaller and provide easier loading of the media into printer **200** as compared to other types of peelers. In other embodiments, rather than include peeler assembly actuator **802**, peeler assembly **704** may be unlatched or otherwise released in response to, e.g., media cover **102** being opened.

(59) FIG. **8A** also shows spindle-less holders **804**, which are adapted to receive media roll **806**, as shown in FIG. **8B**. FIG. **8A** also shows universal printhead **806**. Universal printhead **806** is discussed further in connection with, e.g., FIGS. **148A-14C**. FIG. **8B** also shows peeler assembly **704** as including locking protrusions **808**, which may be used to mate with a latching mechanism to lock peeler assembly **704** in the disengaged, non-peeling position (see, e.g., FIG. **8A**) and/or in the peeling position (see, e.g., FIG. **8C**). For example, locking protrusions **808** can be configured to be engaged by a media cover being shut when the peel bar is in the up or “ready” position as shown in FIG. **8B**. As referred to herein, the “ready” position is between the peeling position and the non-peeling position, such that peeler assembly **704** is ready to engage media units to be peeled from a backing.

(60) Peeler assembly **704** can include gears and other mechanical and/or electrical components that are adapted to automatically engage peeler assembly **704** onto the media as shown in FIG. **8C**. For example, peeler assembly **704** can automatically engage the media in response to media cover **102** being closed while peeler assembly **704** is unlatched. Media cover **102** can be closed either manually and/or electromechanically after the media is loaded into printer **200**. As media cover **102** is closed, peeler assembly **704** can automatically engage the media that has been partially extended between media cover **102** and peeler assembly **704**. As such, peeler assembly **704** is configured to be a threadless peeler, which does not require a label to be partially separated from its backing to enable peeling of subsequent labels. Rather, the peeler assembly may be configured to peel labels after being placed onto a label's printable surface.

(61) For example, media cover **102** can be opened, peeler assembly **704** can be released into an up or other type of ready position as shown in FIG. **8B**, media roll **806** can be loaded into spindle-less holders **804**, media cover **102** can be at least partially closed, a portion of media roll **806** can be at least partially extended between media cover **102** and peeler assembly **704**, and the printing of the labels can cause peeler assembly **704** to peel the labels from their backing. In some embodiments, the peeler assembly can include gears and/or any other mechanism (such as those discussed in connection with FIGS. **9A-10C** and/or **10D-10H**) that enables peeler assembly **704** to be configured to come down and automatically engage media roll **806**, thereby pressing the media against media cover **102**, in response to media cover **102** being closed and/or latched into a closed position. When the media is pressed against the media cover **102**, the media is routed around peel bar **902A** creating a relative sharp bend to cause the peeling to occur as will be discussed further below. Peeler assembly **704** can then separate subsequent media units (e.g., labels) from the media's backing as printer **200** prints. FIG. **8D** shows how peeler assembly **704** may separate media unit **810** from backing portion **812**. The path of the media **812** and the resulting peeling will be further detailed below with respect to FIGS. **10S** and **10T**.

(62) By being configured to threadlessly peel media units from their backing, peeler assembly **704** can be made smaller than other peeler assemblies, since other peeler assemblies designed for manual-use often require user engageable features, such as wings adapted for a user to handle. As such, peeler assembly **704** and other threadless peeler assemblies, some additional examples of which are discussed herein, can lack one or more user engageable features often included in

manually engageable peeler assemblies, such as relatively wider or, more generally, larger peel bars than that shown in connection with peeler assembly **704**.

(63) FIGS. **9A** and **9B** show example peeler assembly **900**, which is another example of a threadless peeler in accordance with some embodiments discussed herein. Peeler assembly **900** includes peel bar **902**, idler bar **902A**, support members **904**, curved slots **906**, actuating shaft **908**, springs **910** and gear **912**.

(64) Components of a printing assembly are also shown in FIG. **9A**. For example,

(65) FIG. **9A** shows roller **914** and printhead **916**. Printhead **916** may be, for example, a universal printhead, such as that discussed in connection with FIGS. **14A** and **14B**.

(66) FIG. **9A** also shows how the printing assembly and peeler assembly **900** can be mounted to the same chassis, namely chassis **918**. Chassis **918**, like other components discussed herein, can be made from any suitable material(s), such as one or more metals, carbon fibers, plastics, rubbers, silicon wafers, among other things. In some embodiments, chassis **918** can also be configured to house and/or protect one or more electrical components, such as a motor, circuitry on a circuit board, among other things.

(67) FIG. **9B** shows peeler assembly **900** without the printing components and chassis **918**. Peel bar **902** can be connected to one or more support members **904A** and **904B** to form a U-shape as shown in FIG. **9B**. In some embodiments, rather than comprise a number of pieces that are connected together, peel bar **902** and one or more of support members **904A** and **904B** can be the same component (e.g., molded or otherwise formed as a single piece of metal). For example, peel bar **902** can be connected to support member **904A** using a hole included in **904A** and a protrusion included at the end of peel bar **902**, and/or peel bar **902** can be the same component as support member **904B** (e.g., a piece of curved metal).

(68) Peel bar **902** can also include curved surface **920** and flat surface **922** separated by edge **924**. Curved surface **920** may be shaped as shown to avoid any sharp and/or pointed edges as well as for aesthetic purposes. Although not shown in FIG. **9B**, one or more rollers, grooves, and/or other features may be included in peel bar **902** (on the side opposite of curved surface **920**) to facilitate peeling a media unit from its backing, while still allowing the media unit to be easily removed from the peel bar and/or while enabling the printer's drive motor to operate more efficiently (than if, e.g., peel bar **902** does not include such features).

(69) Support members **904A** and **904B** are shown in FIG. **9B** as including curved slots **906** through which protrusions **926** of actuating shaft **908** are configured to engage. Protrusions **926** can be located at the distal ends of actuating shaft **908** and be configured to move within curved slots **906**. In some embodiments, peeler assembly **900** may include one or more than two support members. Also, in some embodiments, one or more of the support members may not be configured to engage actuating shaft **908**, or engage actuating shaft **908** in differing manners.

(70) One or more springs **910** may be connected to one or more of support members **904A** and **904B**, and the respective protrusion **926**. Springs **910** can be configured to cause peel bar **902** to be securely stowed when in a non-peeling position and/or be configured to supply a peeling force when peel bar **902** is in the peeling position. In some embodiments, the printer's media cover and/or other component(s) (such as, e.g., a locking component) can be used to securely stow peel bar **902** in either the non-peeling position or the peeling position, or possibly securely stow the peel bar **902** in both the non-peeling position and the peeling position. Using the springs **910** to securely stow the peel bar in both the peeling and non-peeling positions can be accomplished by virtue of the spring being relatively un-extended when the peel bar is in the peeling position and the non-peeling position, and be extended while transitioning between the peeling position and the non-peeling position. Such a configuration would drive the peel bar to the peeling position or the non-peeling position, but not in between.

(71) Peeler assembly **900** may also include one or more gears, such as gear **912**, connected to actuating shaft **908**. Gear **912** may engage actuating shaft **908** (e.g., using one or more teeth or by

any other suitable mechanism). As discussed in connection with FIGS. **10A-10C**, rotating gear **912** may guide peel bar **902** from a non-peeling position (or ready position) to a peeling position (or vice-versa). In some embodiments, gear **912** may be connected to or included in a motor (not shown) configured to drive peel bar to engage/disengage media units being printed. As such, gear **912** and actuating shaft **908** can be configured to move in response to a force exerted by a motor. Gear **912** can also be configured to lock peel bar **902** in a peeling position, ready and/or non-peeling position(s). As another example, peel bar **902**, gear **912** and actuating shaft **908** can be configured to move in response to a force exerted by a user (such as, e.g., a user's finger(s)).

(72) One or more additional components not shown in the drawings may be included in peeler assembly **900**. For example, one or more springs may be attached to (e.g., wound around) a portion of actuating shaft **908** between the support members **904A** and **904B** (e.g., as opposed to springs **910** which are shown as being attached to the defining portion of support members **904A** and **904B**). Such spring may be configured to cause peel bar **902** to assume a ready position (e.g., in response to cover release actuator **124** being depressed) and/or may exert a greater force than springs **910** (e.g., thereby causing springs **910** to be at least partially elongated while in the ready position).

(73) FIGS. **10A-10C** show an example of how peeler assembly **900** can move peel bar **902** from a non-peeling position to a ready position (or vice-versa) and from the ready position to a peeling position (or vice-versa). FIG. **10A** shows peel bar **902** in an example non-peeling position, such as that shown in FIG. **7**. FIG. **10B** shows peel bar **902** in an example ready position, such as that also shown in FIGS. **8A** and **8B**. FIG. **10C** shows peel bar **902** in an example peeling position, such as that also shown in FIGS. **8C** and **8D**.

(74) As noted above, peel bar **902**, gear **912** and actuating shaft **908** can be configured to move in response to, for example, a force exerted by a user onto peel bar **902** and/or a force exerted by an electric motor. While in the non-peeling position, peel bar **902** may be stowed as shown in FIG. **7**. In some embodiments, peel bar **902** may be removed from the non-peeling position to the ready position of FIG. **10B** by a user's finger (e.g., manually lifting the peel bar). In some embodiments, peel bar **902** can move from the non-peeling position of FIG. **10A** to the ready position of FIG. **10B** in response to detecting a depression of a manual release actuator (such as cover release actuator **124**) that unlatches a latch locking peel bar **902** in the non-peeling position and allows a spring to drive peel bar **902** to the ready position of FIG. **10B**. As yet another example, peel bar **902** may move from the non-peeling position to the ready position in response to a button (such as media cover release actuator **124** or any other button) being pushed that actuates a motor.

(75) FIG. **10B** also shows motion arrows **1002**, **1004** and **1006**. Gear **912** may be configured to move in the direction of motion arrow **1002** when peel bar **902** is moved in the direction of motion arrow **1004** from a non-peeling position to a ready position and/or from a ready position to a peeling position. While gear **912** and peel bar **902** are respectively moving in the direction of motion arrows **1002** and **1004**, actuating shaft **908** may move in the direction of motion arrow **1006**. The curved shape of slot **906** can assist in guiding peel bar **902** in the direction of motion arrow **1004**.

(76) As shown in FIG. **10C**, a mechanism, such as spring **910** can assist in and/or independently cause the locking of peel bar **902** in the peeling position. Similar, as shown in FIG. **10A**, a mechanism, such as spring **910** can assist in and/or independently cause the locking of peel bar **902** in the non-peeling position.

(77) In some embodiments, peel bar **902** can be constrained in the non-peeling position, ready position and/or peeling position by one or more other components of the printer. For example, a media cover (and/or peel bar **902**) may be configured to lock the peel bar in the peeling and/or non-peeling position when the media cover is closed. Similarly, in some embodiments, moving of one or more other components of the printer can cause and/or enable peel bar **902** to move among the positions shown in FIGS. **10A-10C**. For example, gear **912** may be configured to automatically

engage the peel bar with the print media in response to the media cover being closed while peel bar **902** is in the ready position of FIG. **10B**. One or more gears, such as gear **912**, can also be configured to lock the peel bar in a peeling position, such as that shown in FIG. **10C**.

(78) FIGS. **10D** and **10E** show example peeler assembly **1010**, which is another example of a threadless peeler in accordance with some embodiments discussed herein. Peeler assembly **1010** may include one or more components that are the same as or similar to those included in peeler assembly **900**. Like reference numbers are used to refer to like components. For example, peeler assembly **1010** includes peel bar **902**, support members **904A** and **904B**, curved slots **906**, actuating shaft **908**, springs **910**, roller **914**, printhead **916** and protrusions **926**, which may be the same or similar as that discussed above.

(79) FIG. **10D** also shows how the printing assembly and peeler assembly **1010** can be mounted to the same chassis, namely chassis **918**. Chassis **918**, like other components discussed herein, can be made from any suitable material(s), such as one or more metals, carbon fibers, plastics, rubbers, silicon wafers, among other things. In some embodiments, chassis **918** can also be configured to house and/or protect one or more electrical components, such as a motor, circuitry on a circuit board, among other things.

(80) FIG. **10E** shows peeler assembly **1010** without the printing components and chassis **918**. Support members **904A** and **904B** are shown in FIG. **10E** as including curved slots **906** through which protrusions **926** of actuating shaft **908** are configured to engage. Protrusions **926** can be located at the distal ends of actuating shaft **908** and be configured to move within curved slots **906**. Actuating shaft **908** can have a rectangular cross-section (as shown in FIG. **10E**), a circular cross-section (as shown in FIG. **9B**) and/or any other suitable shape. Actuating shaft **908** can also include one or more internal gears **1012** located between support members **904A** and **904B** and/or outside of chassis **918**. In some embodiments, peeler assembly **1010** may include one or more than two support members. Also, in some embodiments, one or more of the support members **904A** and **904B** may not be configured to engage actuating shaft **908**, or engage actuating shaft **908** in differing manners.

(81) Peeler assembly **900** may also include one or more gears, such as gear **1014**, connected to internal gears **1012**. Gear **1014** may engage internal gears **1012** and actuating shaft **908** (e.g., using one or more teeth, or a belt, or by any other suitable mechanism). As discussed in connection with FIGS. **10E-10H**, gear **1014** may guide peel bar **902** from a non-peeling position (or ready position) to a peeling position (or vice-versa). In some embodiments, gear **1014** and/or one or more of internal gears **1012** may be connected to slide bar **1016**. Slide bar **1016** may also include handle **1018** configured to drive peel bar **902** to engage/disengage media units being printed in response to, e.g., a user's finger applying pressure in one or more directions. As such, gear **1014**, internal gears **1012** and/or actuating shaft **908** can be configured to move in response to a force exerted by a user and/or anything else that is able to exert a force onto handle **1018**.

(82) One or more additional components not shown in the drawings may be included in peeler assembly **1010**. For example, one or more springs may be attached to (e.g., wound around) a portion of actuating shaft **908** between the support members **904A** and **904B** (e.g., as opposed to springs **910** which are shown as being attached to the defining portion of support members **904A** and **904B**). Such spring may be configured to cause peel bar **902** to assume a ready position (e.g., in response to cover release actuator **124** being depressed) and/or may exert a greater force than springs **910** (e.g., thereby causing springs **910** to be at least partially elongated while in the ready position).

(83) FIGS. **10F-10H** show an example of how peeler assembly **1010** can move peel bar **902** from a non-peeling position to a ready position (or vice-versa) and from the ready position to a peeling position (or vice-versa). FIG. **10F** shows peel bar **902** of peeler assembly **1010** in an example non-peeling position, such as that shown in FIG. **7**. FIG. **10G** shows peel bar **902** of peeler assembly **1010** in an example ready position, such as that also shown in FIGS. **8A** and **8B**. FIG. **10C** shows



peel bar **902** of peeling assembly **1010** in an example peeling position, such as that also shown in FIGS. **8C** and **8D**. While some example embodiments described herein provide for a peel bar that may be disposed in a non-peeling, ready, and peeling position, other embodiments may provide only for the peel bar to be in either a peeling or non-peeling position, where the peel bar is moved between the two positions. In such an embodiment, the “ready” position of illustrated embodiments, such as FIG. **10G**, may only be a position of the peel bar as it is moved between the non-peeling and peeling positions.

(84) As noted above, peel bar **902**, gear **1014** and actuating shaft **908** can be configured to move in response to, for example, a force exerted by a user's finger onto handle **1018**. While in the non-peeling position, peel bar **902** may be stowed as shown in FIG. **10F**. In some embodiments, peel bar **902** may be removed from the non-peeling position to the ready position of FIG. **10G** by a user's finger (e.g., pushing handle **1018** in the direction of motion arrow **1020**).

(85) FIG. **10G** also shows motion arrows **1022** and **1024**. Gear **1014** may be configured to move in the direction of motion arrow **1022** when peel bar **902** is moved in the direction of motion arrow **1024** from a non-peeling position to a ready position and/or from a ready position to a peeling position. While gear **1014** and peel bar **902** are respectively moving in the direction of motion arrows **1022** and **1024**, slide bar **1016** may move in the direction of motion arrow **1018**. The curved shape of slot **906** can assist in guiding peel bar **902** in the direction of motion arrow **1024**.

(86) As shown in FIG. **10H**, a mechanism, such as spring **910** can assist in and/or independently cause the locking of peel bar **902** in the peeling position. Similar, as shown in FIG. **10A**, a mechanism, such as spring **910** can assist in and/or independently cause the locking of peel bar **902** in the non-peeling position.

(87) FIGS. **10I-10L** show example peeler assembly **1030**, which is another example of a threadless peeler in accordance with some embodiments discussed herein. Peeler assembly **1030** may include one or more components mounted to a printer's frame, such as chassis **918**, and a specially designed media cover **102**.

(88) For example, media cover **102** of peeler assembly **1030** may include cam slot **1032** and idler bar **1034**. Media cover **102** may also include hole **1036** (which need not be a hole and may be a detent, recess and/or any other suitable means) for receiving spring loaded plunger **1038**. Spring loaded plunger **1038** may be included and/or otherwise attached to chassis **918**, the printer's housing and/or any other component of the printer. Spring loaded plunger **1038** may be retracted from hole **1036** in response to, for example, cover release actuator **124** being depressed and/or in response to any other type of user interaction and/or automated control signal being generated and/or received by the printer.

(89) Peeler assembly **1030** may also include peeler bar **1040**, torsion bar **1042**, and latch **1044**. Peeler **1040** may function and/or otherwise be the same as or similar to peel bar **902** discussed above. For example, peeler **1040** may include rollers on its peeling surface as described herein. Peeler **1040** or torsion bar **1042** may also be linked via, e.g., protrusion **1046** and a slot included in one or more support members of peeler **1040**.

(90) Latch **1044** may include spring **1048**, which may be configured to store enough potential energy to cause latch **1044** to stay engaged with torsion bar **1042**, despite the force applied to torsion bar **1042** by spring **1050**. Spring **1048** may also be configured to enable a user to depress latch **1044** to release peeler **1040**.

(91) For example, as shown in FIG. **10J**, while media cover **102** is open a user may apply a force in the direction of motion arrow **1056**, which will cause latch **1044** to disengage torsion bar **1042**. Upon being disengaged, spring **1050** can cause torsion bar to rotate in the direction of motion arrow **1058**. The portion of torsion bar **1042** that was engaged with latch **1044** can then move in the direction of motion arrow **1060** as guided by protrusion **1046** in slot **1052** of the support member of peeler **1040**. As a result, peeler **1040** may move in the direction of motion arrow **1062** and enter a ready position. Upon media cover **102** being closed in the direction of motion arrow **1064**, one or

more cams **1054** (only one is shown in the view of the drawings) can be engaged by one or more cam slots **1032** (only one can be seen in the drawings) and peeler **1040** can be locked into a peeling position. In some embodiments, a user may have to extend a media unit on a backing into space **1066**, which is located between peeler **1040** and media cover **102**, before closing media cover **102**. (92) Also shown in FIG. **10I** is circuitry **1068**, which may be configured to execute some or all of the printer's functionality, some examples of which are discussed herein in connection with, e.g., FIG. **11B**.

(93) FIG. **10K** shows peeler **1040** in a peeling position. As a media unit on a backing moves over idler bar **1034** and under peeler **1040**, the media unit can be peeled from the backing and can stick to or otherwise be removed from defining portion of peeler **1040** that is visible in FIG. **10K**. The peeling process is further described with respect to FIGS. **10S** and **10T** below.

(94) FIG. **10L** shows a left side view of peeler assembly **1030** in a stowed position (as compared to the right side view of peeler assembly **1030** in the stowed position shown in FIG. **10I**). From the view shown in FIG. **10L**, switch **1070** is visible. Switch **1070** can be configured to detect if peeler assembly **1030** is in the stowed position of FIGS. **10I** and **10L**, or in the peeling position of FIG. **10K** and/or the ready position of FIG. **10J**. Switch **1070** can be any suitable component, such as one or more of a pressure sensitive switch, optical switch, any other type of proximity switch, etc. In some embodiments, a sensor could determine the status of the switch and communicate that to the processor which would then determine the position of the peeler. In another embodiment, a user could use a key or the touch screen to indicate the desire to turn peel on or off, and the printer could then move the peel bar from one position to another. The position of the peeler may impact printer settings such that a sensor to detect the position of the peel bar may cause the printer settings to be adjusted to correspond with the peel bar position as described further below. FIG. **10M** illustrates another example embodiment of a peeler assembly **1500** according to an example embodiment of the present invention. The illustrated peeler assembly **1500** includes peel bar **902** and peel bar support member **904**. The peel bar **902** is attached to the peel bar support member **904**, and the peel bar support is coupled to the chassis at pivot point **1505**. The peeler assembly **1500** of FIG. **10M** includes an actuating shaft **908** extending through the chassis **918**, and coupled to support members **904** on either side of the chassis **918** by protrusions **926** within curved slots **906**. While the illustrated side view of the peeler assembly **1500** including the support members **904**, pivot points **1505**, actuating shaft **908**, protrusions **926**, and curved slots **906** are substantially mirrored on the opposite side of the chassis **918**, the mechanism by which the peel bar **902** is actuated from a peeling position to a stowed, non-peeling position may be disposed only on one side of the chassis **918**.

(95) The mechanism for actuating the peel bar **902** of the peeler assembly **1500** from the illustrated stowed, non-peeling position of FIG. **10M** includes a pinion gear **1510**, a peeler assembly actuator **1515**, a rack **1520** attached to the peeler assembly actuator **1515** which engages the pinion gear **1510**, and a slot **1525** in the chassis **918** in which the peeler assembly actuator **1515** is configured to travel. The peel bar **902** is illustrated disposed in the stowed, non-peeling position while the peeler assembly actuator **1515** is disposed in the disengaged position at the bottom of its travel. In response to the peeler assembly actuator being pressed or actuated in the direction of arrow **1530**, the peeler assembly actuator **1515** moves along slot **1525** (to which it is engaged by, for example, fastener **1535**) toward an engaged position. As the peeler assembly actuator **1515** advances along the direction of arrow **1530**, the rack **1520** moves relative to pinion gear **1510** and rotates pinion gear **1510** in a clockwise direction. As pinion gear **1510** rotates in a clockwise direction, the actuating shaft **908**, which is engaged with the pinion gear **1510** by a mating gear (not shown) is rotated in a counter-clockwise direction. Rotation of the actuating shaft **908** in the counter-clockwise direction moves protrusion **926** within curved slot **906** and pivots the peel support member **904** about pivot point **1505**. The peel bar **902** rotates in a clockwise direction around the pivot point from the stowed, non-peeling position of FIG. **10M**, to the deployed, peeling position of

FIG. 10N.

(96) Example embodiments may include a spring member **1540** which biases the peeler assembly actuator **1515** in the direction of arrow **1545**, toward the disengaged position, thereby biasing the peeler assembly **1500** in the stowed, non-peeling position of FIG. **10M**. However, the peeler assembly **1500** may be retained in the deployed, peeling position by springs, such as those shown (springs **910**) in FIGS. **10F** through **10H**. A locking mechanism may also be employed to maintain the peeler assembly in the peeling position and/or the peeler assembly actuator in the engaged position, as will be detailed further below.

(97) The movement of the peeler assembly **1500** of FIGS. **10M** and **10N** from the stowed, non-peeling position of FIG. **10M** to the deployed, peeling position of FIG. **10N**, and vice versa, may operate in generally the same manner as the peel bar **902** and assembly of FIGS. **10F-10H**; however, the peeler assembly actuator **1515** and the attached rack **1520** are configured to move in a generally vertical direction (along arrow **1530** of FIG. **10M**) as compared to a generally horizontal direction (along arrow **1020** of FIG. **10G**). The peeler assembly of the embodiment of FIGS. **10M-10R** may not engage or otherwise rely upon closure of a media cover to secure the peeler assembly in the peeling position, as described with respect to the embodiment of FIGS. **10I-10L**.

(98) FIG. **10P** illustrates a side-view of a printer which may include a peeler assembly and associated actuation mechanism as illustrated in FIGS. **10M** and **10N**. In the illustrated embodiment, the printer **1600** includes a housing **1610**, a media cover **1620**, a media cover release actuator **1630**, and peeler assembly actuator **1640**. Both the peeler assembly actuator **1640** and the media cover release actuator **1630** are accessible when the media cover **1620** is in the closed position. The peeler assembly actuator **1640** further includes lock button **1670** as will be described further below. The peeler assembly of the illustrated embodiment is actuated from the stowed, non-peeling position to the deployed, peeling position by advancing the peeler assembly actuator **1640** in the direction of arrow **1645**, from the illustrated disengaged position corresponding to a stowed, non-peeling position of the peeler assembly of FIG. **10P** to an engaged position corresponding to a deployed, peeling position of the peeler assembly of FIG. **10Q**. It may be desirable for the peeler assembly actuator **1640** to be accessible outside of the housing **1600** when the media cover **1620** is in the closed position such that previously loaded media may not require threading between the peel bar during the media loading process. Instead, the media may be guided on the proper path by a simple loading technique (not requiring threading of the media) and engagement of the peeler mechanism after the media cover is closed, as further described below.

(99) The media cover release actuator **1630** may be depressed in a direction opposite to that of arrow **1645** of FIG. **10P** in order to release and open the media cover **1620** for the loading or unloading of media. However, it may be undesirable to release and/or open the media cover **1620** while the peel bar **902** is in the deployed, peeling position. The media cover **1620** may not properly open or close with the peel bar in the peeling position. While described herein as a media cover release actuator to release the media cover from the closed position, embodiments of the media cover release actuator may also actuate a media cover open mechanism that drives the media cover to the open position by electro-mechanical means, spring biasing means, hydraulic/pneumatic means, gravity, etc. As such, the term “release” as used herein with reference to the media cover release actuator refers to transitioning the media cover from the closed position, at least in part, toward the open position.

(100) As shown, in FIG. **10P**, the media cover release actuator **1630** has clearance to be moved in a direction opposite to that of arrow **1645**, with the clearance illustrated as **1655** in order to release and open the media cover **1620**. As shown in FIG. **10Q**, with the peeler assembly actuator **1640** in the engaged position corresponding to the deployed, peeling position of the peeler assembly, the media cover release actuator **1630** does not have sufficient clearance **1655** to be depressed.

Therefore, with the peeler assembly actuator **1640** in the engaged position corresponding to the peeler assembly being in the deployed, peeling position, the media cover release actuator **1630**

cannot release the media cover **1620** from the closed position. This feature prevents the media cover from being released from the closed position while the peeler assembly is in the peeling position, as opening of the media cover while the peel bar is in the peeling position can introduce error states, such as media jamming and interference between the media cover and the peel bar, precluding full travel of the media cover or closure of the media cover.

(101) While printing with printers of example embodiments of the present invention, it may be undesirable for the peel bar to be accidentally or unintentionally moved from the peeling position to the non-peeling position. To that end, FIGS. **10P**, **10Q**, and **10R** each illustrate a lock button **1670** configured to preclude movement of the peeler assembly actuator **1640** from the engaged position corresponding to the deployed, peeling position of the peel bar to the disengaged position corresponding to the stowed, non-peeling position of the peel bar.

(102) As the spring **1540** of FIG. **10N** biases the peeler assembly actuator **1640** to the disengaged position corresponding to the stowed, non-peeling position of the peel bar, it may be desirable to latch or lock the peeler assembly actuator **1640** in the raised, engaged position of FIG. **10Q**. Inadvertent movement of the peel bar **902** from the peeling position (FIG. **10M**) to the non-peeling position (FIG. **10N**) during printing may cause the media to become jammed in the printer and may lead to printing errors. Therefore, a lock mechanism including lock button **1670** may be implemented to preclude inadvertent movement of the peeler assembly actuator **1640**. When the peeler assembly actuator **1640** is in the engaged position of FIG. **10Q** corresponding to peeling, a user may need to depress the lock button **1670** before moving the peeler assembly actuator **1640** to the disengaged, non-peeling position. Thus discouraging accidental movement of the actuator and preventing accidental bumping of the actuator **1640** to the disengaged position. When the peeler assembly actuator **1640** is in the engaged position illustrated in FIG. **10Q**, the locking mechanism including the lock button **1670** is in the locked position such that the peeler assembly actuator **1640** cannot be accidentally “bumped” back to the disengaged position illustrated in FIG. **10P**.

(103) As outlined above, in some embodiments of a printer including a media cover **1620** and a peeler assembly, it may be detrimental to open the media cover **1620** while the peeler assembly is disposed in a deployed, peeling position. A corollary of this may be that it may also be detrimental for the peeler assembly to be moved from the stowed, non-peeling position to the deployed, peeling position while the media cover **1620** is in the open position. FIG. **10R** illustrates the printer of FIGS. **10M** and **10P** with the media cover **1620** disposed in the open position and the peeler assembly actuator **1640** in the position corresponding to the stowed, non-peeling position of the peeler assembly.

(104) In some example embodiments, the lock button **1670** may also be implemented to also prevent movement of the peeler assembly actuator from the disengaged, non-peeling position to the engaged, peeling position when the media cover **1620** is in the open position. When the media cover **1620** is in the closed position, as shown in FIGS. **10P** and **10Q**, the peeler assembly actuator **1640** may be moved along arrow **1645** of FIG. **10P** upon pressing of lock button **1670**. The depression of lock button **1670** may allow the peeler assembly actuator **1640** to be advanced along arrow **1645**. However, when the media cover **1620** is in the open position, as illustrated in FIG. **10R**, lock button **1670** may be precluded from being depressed by a lock mechanism and the peeler actuation actuator **1640** is precluded from moving from the disengaged position corresponding to the stowed, non-peeling position. This lock-out feature may prevent the peel bar from being deployed to the peeling position while the media cover **1620** is open, as when the peel bar is in the deployed position, the media cover may not be able to properly close. This lock-out feature may dissuade a user from attempting to close the media cover **1620** with the peel bar deployed.

(105) As outlined above with regard to FIGS. **10P**, **10Q**, and **10R**, example embodiments of the invention may be configured to preclude a user from improperly deploying the peeler assembly and improperly opening the media cover. Precluding a user from improper use of the printer may lead to fewer problems with the printer and increased user satisfaction.

(106) An example embodiment demonstrating advantages of the peeler assembly actuator **1640** being positioned to be accessible outside of the housing may be described with respect to the loading of media into the printer. FIG. **10R** illustrates a printer **1600** with the media cover **1620** in an open position. As outlined above, when the media cover is in the open position, it may be undesirable for the peeler assembly to be in the peeling position such that the peeling assembly of the illustrated embodiment of FIG. **10R** may be in the non-peeling position. If the peeler was disposed in the peeling position, loading of the media would require threading of the media around an idler bar (described and illustrated below) and under the peeler bar. This threading operation can be cumbersome, particularly in a compact printer.

(107) Upon loading a roll of media, for example including a roll labels disposed on a backing, the media cover **1620** may be closed. FIG. **10S** illustrates a cross section of a portion of a printer similar to those embodiments illustrated in FIGS. **10A-10R** in which a media roll **1680** is received within the printer. The illustrated embodiment includes the peel bar **902**, the peel bar supports **904**, the platen roller **907**, the printhead **903**, and the media roll **1680**. The illustrated embodiment further includes a tear bar **1685**. In the illustrated embodiment, the platen roller **907** may be attached to the media cover **1620**. A strip of media (and media backing) may be separated from the roll as the media roll **1680** is loaded into the printer, and upon closing the media cover **1620**, the platen roller **907** sandwiches the strip of media between the printhead **903** and the platen roller **907** at printing nip **903A**. The path of the media is illustrated by arrow **901**.

(108) The illustrated embodiment of FIG. **10S** depicts a strip of media extending from the media roll **1680** between the printhead **903** and the platen roller **907**, and exiting the printer along the media path shown by arrow **901**. Media that exits the printer in this manner may be configured for tearing at intervals to separate printed media from the strip of media. Tear bar **1685** may be a serrated or otherwise sharpened edge along which the media may be torn. When printing to media that is configured to be torn, or linerless media which does not require the printed substrate to be separated from a backing, the printed media may exit the printing nip **903A** toward the tear bar **1685**. With the media positioned against the tear bar **1685**, tearing of the media is more precise.

(109) FIG. **10T** illustrates an example embodiment of the cross section of the printer as shown in FIG. **10S** printing to media that is attached to a backing. The media roll **1680** may be received in the same manner as described with respect to FIG. **10S**. However, after the media is loaded, the peel bar supports **904** may be pivoted as shown by arrow **1690** to move the peel bar **902** to the peeling position. When the peel bar is moved to the peeling position shown in FIG. **10T**, the media may be folded around idler bar **902A** and under peel bar **902**. This positioning of the media does not require manual intervention other than to move the peel bar from the non-peeling position to the peeling position. Further, as noted above, the movement of the peel bar may be automated such that manual actuation isn't necessary. The strip of media is then disposed in the media backing feed path illustrated by arrow **905** in FIG. **10T**. As the media is advanced during printing, the relatively sharp contours of the media backing feed path **905**, particularly the sharp contour around the idler bar **902A** as held by the peel bar **902**, cause the media to be separated or peeled from the backing. The media advances along media path **909** while the backing advances along media backing path **905**. Due to the sharp contour around idler bar **902A**, the relatively high stiffness of the media, and the relatively low adhesive force between the media and the media backing, the media advances along media path **909** while the media backing advances along media backing path **905** without requiring manual peeling of the media from the backing.

(110) The media path **909** of FIG. **10T** exits the printing nip **903A** in a direction different than that of the media path **901** of FIG. **10S**. As such, the printing settings of may need to be adjusted in order to optimize print quality. Printing settings may be adjusted in response to the position of the peel bar in order to optimize printing. Printhead settings, such as printing temperature, printing speed, print character position, etc., may be adjusted based on the peel bar position to improve the printing quality while the speed of the printing and media feeding may also be adjusted to optimize

print quality.

(111) As described above with respect to FIG. **10L**, a switch **1070** may be used to provide the printer software with an indication of the peeler position. The software of the printer may receive this signal and adjust the printer settings, such as the printhead settings, accordingly.

(112) FIG. **11A** shows printer **1100**, which may include one or more components that are the same and/or similar to the other printers discussed herein. In some embodiments, one or more sensors, such as sensor **1102** of FIG. **11A**, can be configured to detect whether peel bar **902** is in a peeling position. For example, sensor **1102** may be an optical, magnetic, pressure (e.g., depression) and/or proximity sensor that is able to detect when peel bar **902** is in a peeling position. In other embodiments, one or more other sensors may be alternatively or additionally included in printer **1100**. For example, one or more sensors may be used to determine the position of a support member, such as support member **904A**. As another example, one or more sensors may be included inside of the printer housing (e.g., on the frame of the printer, on a circuit board and/or at any other suitable location).

(113) FIG. **11B** shows a block diagram of example circuitry **1068** that may be included in a printer. As shown in FIG. **11B**, in accordance with some embodiments, circuitry **1068** includes various means, such as processor **1104**, memory **1106**, communication interface **1108** and user interface **1110** that can be configured to perform the various functions herein described. These means of circuitry **1068** as described herein may be embodied as, for example, hardware elements, including control circuitry (e.g., processor **1104**, including any suitably programmed processor and/or combinational logic circuit, among other things), a computer program product comprising computer-readable program instructions (e.g., software/firmware) stored on a nontransitory computer-readable medium (e.g., memory **1106**) that is executable by the printer's other circuitry (e.g., processor **1104**), or some combination thereof.

(114) Processor **1104** may, for example, be embodied as various means including one or more microprocessors with accompanying digital signal processor(s), processor(s) without an accompanying digital signal processor, coprocessors, multi-core processors, controllers, computers, various other processing elements including integrated circuits such as, for example, an ASIC (application specific integrated circuit) or FPGA (field programmable gate array), or some combination thereof. Accordingly, although shown in FIG. **11B** as a single processor, in some embodiments processor **1104** comprises a plurality of processors and/or any other type of control circuitry. The plurality of processors, for example, may be embodied on a single computing device or may be distributed across a plurality of computing devices collectively configured to function as a printer. The plurality of processors may be in operative communication with each other and may be collectively configured to perform one or more functionalities of circuitry **1068** as described herein. In an example embodiment, processor **1104** is configured to execute instructions stored in memory **1106** and/or that are otherwise accessible to processor **1104**. These instructions, when executed by processor **1104**, may cause the printer to perform one or more of the functionalities as described herein. As such, whether configured by hardware, firmware/software methods, or by a combination thereof, processor **1104** may comprise an entity capable of performing operations according to embodiments of the present invention while configured accordingly. Thus, for example, when processor **1104** is embodied as an ASIC, FPGA or the like, processor **1104** may comprise specifically configured hardware for conducting one or more operations described herein. Alternatively, as another example, when processor **1104** is embodied as an executor of instructions, such as may be stored in memory **1106**, the instructions may specifically configure processor **1104** to perform one or more algorithms and operations described herein.

(115) Memory **1106** may comprise, for example, volatile storage, non-volatile storage, or some combination thereof. Although shown in FIG. **11B** as a single memory component, memory **1106** may comprise a plurality of memory components. The plurality of memory components may be embodied on a single computing device or distributed across a plurality of computing devices. In

various embodiments, memory **1106** may comprise, for example, a hard disk, random access memory, cache memory, flash memory, a compact disc read only memory (CD-ROM), digital versatile disc read only memory (DVD-ROM), circuitry configured to store information, any other type of memory, or some combination thereof. Memory **1106** may be configured to store information, data, applications and instructions, among other things, for enabling circuitry **1068** to provide various functionality in accordance with some example embodiments of the present invention. For example, memory **1106** can be configured to buffer input data for processing by processor **1104**. Additionally or alternatively, in at least some embodiments, memory **1106** is configured to store program instructions for execution by processor **1104**. Memory **1106** may store information in the form of static and/or dynamic information. This stored information may be stored and/or used by circuitry **1068** during the course of performing its functions.

(116) Communication interface **1108** may be embodied as any device or means embodied in circuitry, hardware, a computer program product comprising computer readable program instructions stored on a computer readable medium (e.g., memory **1106**) and executed by a processing device (e.g., processor **1104**), or a combination thereof that is configured to receive and/or transmit data from/to another device, such as, for example, a second printer and/or the like. In at least one embodiment, communication interface **1108** is at least partially embodied as or otherwise controlled by processor **1104**. In this regard, communication interface **1108** may be in communication with processor **1104**, such as via a bus (not shown). Communication interface **1108** may include, for example, an antenna, a transmitter, a receiver, a transceiver, network interface card and/or supporting hardware and/or firmware/software for enabling communications with another computing device. Communications interface **1108** may also include one or more sensors, such as sensor **1102**, which can be configured to detect the position of peel bar **902** (e.g., peeling position, ready position and/or non-peeling position), among other things. Communication interface **1108** may also be configured to receive and/or transmit data using any protocol suitable for facilitating communications between computing and/or other types of devices. Exemplary docking stations are provided in commonly-assigned U.S. patent application Ser. No. 13/085,431, titled “MOBILE PRINTER NETWORKING AND INTERFACING,” which was incorporated herein in its entirety by reference. Communication interface **1108** may additionally be in communication with the memory **1106**, user interface **1110** and/or any other component of the printer, such as via a bus (not shown).

(117) User interface **1110** may be in communication with processor **1104** to receive an indication of a user input and/or to provide an audible, visual, mechanical, or other output to a user. As such, user interface **1110** may include, for example, display **108**, left button **110**, right button **112**, menu button **114**, back button **116**, navigation buttons **118**, feed button **120**, power button **122**, an audio transducer, and/or other input/output mechanisms.

(118) Circuitry **1068** may also include one or more drive motors **1112**, which may be configured to move one or more media units when printing and/or to move peel bar **902** from and/or to one or more positions, among other things. For example, sensor **1102** may be configured to provide an indication of the position of the peel bar. Processor **1104** (and/or any other type of control circuitry) can be configured to receive the peel bar position indication from sensor **1102**. In response to an indication representing peel bar **902** being in a non-peeling position, for example, processor **1104** can be configured to retrieve a drive motor parameter and cause electric drive motor **1112** to move print media relative to a printhead at a first speed and/or in accordance one or more other drive motor parameters stored in memory (such as at a predetermined torque). In response to an indication representing peel bar **902** being in a peeling position, for example, processor **1104** can be configured to retrieve a drive motor parameter and cause electric drive motor **1112** to move the print media relative to the printhead at a second speed and/or in accordance one or more other drive motor parameters stored in memory (such as at a greater torque). The first and second speeds and/or drive motor parameters retrieved can be different. For example, the second speed can be

slower than the first speed, or vice-versa. As another example, the torque may be increased when the peel bar is in a peeling position. As another example, the optimum position of the media unit with respect to the printhead may be adjusted, either while printing (position of the printed dots relative to the edge of a label) or after printing (position of the printed label relative to the printhead, tear bar, or peel sensor) or both.

(119) Memory **1106** can be configured to store a drive table that includes drive motor parameters and/or other types of instructions executable by the control circuitry. The drive table's instructions can include, for example, parameters associated with the first speed and the second speed associated with, e.g., the position of the peel bar. As a further example, processor **1104** can be further configured to cause electric drive motor **1112** to move the print media relative to the printhead at the second speed in response to determining the printer is experiencing or is likely to experience an overheating error.

(120) In addition to or instead of adjusting the print speed based on the position of the peel bar (and/or any other sensor indication, such as battery power), processor **1104** can be configured to enter a non-peel or peel mode that includes other parameters. For example, while in the peel mode a display (e.g., icon, words, etc.) may be presented indicating that the peel bar is in a peeling position. Similar displays may be presented for other modes (based on, e.g., positions of the peel bar). In some embodiments, for example, the lack of an icon and/or other display may be used to inform the user of the printer's operational mode. For example, the lack of a peel mode indication indicator (e.g., icon) being displayed may indicate to the user that the printer is in non-peel mode.

(121) As such, some embodiments discussed herein can reduce the possibility that the label media may become misaligned, resulting in a wasted label. In some embodiments, there is no need to open a latch or cover. The user can switch to peel mode and back to non-peel mode as often as desired without wasting labels.

(122) FIG. **12** shows an exemplary method, namely process **1200**, that may be implemented by and/or with the processor and/or other components of a printer, such as mobile printer **100** and/or printer **200**. Like some other processes discussed herein, process **1200** is represented by a flow diagram in accordance with some exemplary methods, computer program products and/or systems discussed herein, including printers **100**, **200** and **1100**. It will be understood that each operation, action, step and/or other types of functions shown in the diagram, and/or combinations of functions in the diagrams, can be implemented by various means. Means for implementing the functions of the flow diagram, combinations of the actions in the diagrams, and/or other functionality of example embodiments of the present invention described herein, may include hardware and/or a computer program product including a computer-readable storage medium (as opposed to or in addition to a computer-readable transmission medium) having one or more computer program code instructions, program instructions, or executable computer-readable program code instructions stored therein. For example, program code instructions associated with FIG. **12** may be stored on one or more storage devices, such as memory **1106**, and executed by one or more processors, such as processor **1104**. Additionally or alternatively, one or more of the program code instructions discussed herein may be stored and/or performed by distributed components, such as those discussed in connection with printers **100**, **200** and **1100**. As will be appreciated, any such program code instructions may be loaded onto computers, processors, other programmable apparatuses (e.g., printer **100**, **200** or **1100**) from one or more computer-readable storage mediums (e.g., memory **1106**) to produce a particular machine, such that the particular machine becomes a means for implementing the functions of the actions discussed in connection with, e.g., FIG. **12** and/or the other drawings discussed herein.

(123) The program code instructions stored on the programmable apparatus may also be stored in a nontransitory computer-readable storage medium that can direct a computer, a processor (such as processor **1104**) and/or other programmable apparatus to function in a particular manner to thereby generate a particular article of manufacture. The article of manufacture becomes a means for



implementing the functions of the actions discussed in connection with, e.g., FIG. 12 and the other flow chart included herein. The program code instructions may be retrieved from a computer-readable storage medium and loaded into a computer, processor, or other programmable apparatus to configure the computer, processor, or other programmable apparatus to execute actions to be performed on or by the computer, processor, or other programmable apparatus. Retrieval, loading, and execution of the program code instructions may be performed sequentially such that one instruction is retrieved, loaded, and executed at a time. In some example embodiments, retrieval, loading and/or execution may be performed in parallel by one or more machines, such that multiple instructions are retrieved, loaded, and/or executed together. Execution of the program code instructions may produce a computer-implemented process such that the instructions executed by the computer, processor, other programmable apparatus, or network thereof provides actions for implementing the functions specified in the actions discussed in connection with, e.g., process **1200** of FIG. 12.

(124) Process **1200** starts at **1202** and at **1204** the printer receives an indication of a user's desire to open the printer's media cover. For example, the user may touch a virtual button presented by a touch-display, an electromechanical button, a remote control button, a mechanical button (such as cover release actuator **124**), and/or utilize any other means for indicating a desire to open the printer's media cover.

(125) In response to receiving the peel bar position indication of the user's desire to open the media cover, the printer may, at **1206**, unlatch or otherwise open the media cover. At **1208**, the printer receives media for printing and/or encoding. For example, the printer may receive paper, a roll of labels on a backing, and/or any other type of media onto which indicia can be printed and/or programmed.

(126) If the media received at **1208** is attached to a liner or other backing, the media may be peeled from the backing by a peeler after printing. If a peeler bar is to be used, **1212** can be next in process **1200**.

(127) At **1212**, the printer receives an indication of a user's desire to release the printer's peeler bar from its latched position. For example, a user may depress a push button, such as push button **802**. The printer can be adapted to release its peeler bar in response to receiving the peel bar position indication of a user's desire to release the peeler bar. In other embodiments, the peeler bar may also or instead be released in response to the printer's circuitry determining that the peeler bar should be released. For example, the media loaded into the printer may include a radio frequency tag, visual indicia (e.g., marks on the backing), any other means, or combination thereof that includes data and/or that is otherwise interpreted by the printer to mean the media is to be peeled from a backing after printing. In some embodiments, printer **400** may include a spindle that includes a radio frequency identification ("RFID") reader, some examples of which are discussed in commonly-assigned U.S. Patent Application Publication No. 2008-0298870, titled "MEDIA PROCESSING SYSTEM AND ASSOCIATED SPINDLE," which is hereby incorporated by reference in its entirety.

(128) At **1216**, the printer prints/encodes the media with the peeler bar engaged. The printer can also be configured to monitor for error conditions while printing. For example, at **1218** the printer determines whether the media needs to be replenished (e.g., is there enough media for the remaining or expected print job). In response to determining that the media needs replenishing, process **1200** returns to **1204** and the printer can wait to receive an indication of the user's desire to open the media cover. In some embodiments, process **1200** may return to **1206** and the media cover may be opened automatically by the printer. The printer may also or instead display a message on its display screen and/or on a remote display screen that the printer is low or out of media and needs replenishing.

(129) In response to determining at **1218** that the media does not need replenishing, process **1200** can proceed to **1220** and the printer determines whether more indicia needs to be printed/encoded.

For example, the printer may have received an initial print command for a print job involving the printing of a number of labels, which has not yet been completed. If more labels are to be printed/encoded, process **1200** returns to **1216** and the printer prints the media with the peeler bar engaged.

(130) At **1222**, the printer can be configured to wait for another print command before printing/encoding more indicia. The printer may wait at **1222** if, for example, the previous print job has been completed and/or a sensor is detecting the peeled label affixed to the peeler bar, among other reasons.

(131) While waiting for a print command at **1222** (or at any other time), a user may decide and/or the printer may be configured to decide stop using the peel bar at **1224**. In response to a determination to continue to use the peel bar, process **1200** can proceed to **1226**.

(132) At **1226**, the printer may determine whether it should shut down, enter a sleep mode (due to, e.g., a predetermined time period lapsing, the power button being depressed, etc.), and/or enter a partial sleep mode (shutting down or otherwise reducing the power draw of one or more components). If the printer determines it should not shut down or enter a sleep mode, process **1200** can return to **1220**. If the printer determines at **1226** the printer should enter a power down or sleep mode, process **1200** ends at **1228**.

(133) Returning to **1210**, in response to determining that the print job is to be executed without the peeler bar, process **1200** proceeds to **1230**. At **1230**, the peeler bar remains latched in the closed position and at **1232** the media is printed by the printer without using the peeler bar. Process **1200** may also proceed to **1232** in response to the peel bar being stowed at **1224**.

(134) At **1234**, the printer can be configured to determine whether the media needs to be replenished (e.g., is there enough media for the remaining or expected print job). In response to determining that the media needs replenishing, process **1200** returns to **1204** and the printer can wait to receive an indication of the user's desire to open the media cover. In some embodiments, process **1200** may return to **1206** and the media cover may be opened automatically by the printer, unlocked for a user to open, etc. The printer may also or instead display a message on its display screen and/or on a remote display screen that the printer is low or out of media and needs replenishing.

(135) In response to determining at **1234** that the media does not need replenishing, process **1200** proceeds to **1236** and the printer can determine whether more indicia needs to be printed/encoded. For example, the printer may have received an initial print command for a print job involving the printing of a number of labels which has not yet been completed. If more sheets are to be printed/encoded, process **1200** returns to **1232** and the printer prints the media with the peeler bar latched or otherwise disengaged from the printing process.

(136) At **1238**, the printer can be configured to wait for another print command before printing/encoding more indicia. The printer may wait at **1238** if, for example, the previous print job has been completed, a sensor is detecting an error condition (such as overheating), among other reasons.

(137) While waiting for a print command at **1238**, a user may decide and/or the printer may be configured to decide start using the peel bar at **1240**. In response to a determination to use the peel bar, process **1200** can proceed to **1214**.

(138) In response to a determination to continue not using the peel bar (e.g., allow the peel bar to remain in the non-peeling position), process **1200** can proceed to **1242**.

(139) The printer may then determine at **1242** whether it should shut down or enter a sleep mode (due to, e.g., a predetermined time period lapsing, the power button being depressed, etc.). If the printer determines it should not shut down or enter a sleep mode, process **1200** returns to **1236**. If the printer determines at **1242** the printer should enter a power down or sleep mode, process **1200** ends at **1228**.

(140) FIG. 13 shows process **1300** that can be implemented using a printer's processor and/or other

components in accordance with some embodiments. Process **1300** can enable a printer to determine whether the peeler bar is engaged or disengaged, and dynamically adjust the print speed accordingly.

(141) For example, the printer's circuitry can communicate with at least one peeler sensor that monitors whether the peeler bar is engaged. The peeler sensor(s) can be incorporated on the printer's circuit board, in the media cover, within the printer's housing, and/or elsewhere within the printer. In response to the peeler sensor indicating the peeler is engaged, the printer's control circuitry can be configured to provide relatively more electrical current to the print feed motor. Dynamically adjusting the current of the drive motor may improve print quality (with or without changing the print speed) by providing more torque when a peeler bar, such as peeler **704**, is engaged. The additional torque can compensate for the drag applied to the print media by the peeler bar. Similarly, relatively less electrical current can be provided to the drive motor to reduce the torque when the peeler bar is not engaged. By providing less current when the peeler bar is disengaged, the printer's battery life can be extended without sacrificing print quality. The amount of electrical current can be determined based on, for example, a drive motor parameter and/or any other type of instructions received and/or retrieved by the printer's processor. Process **1300** may benefit, among other things, a portable printer that may be relatively limited in available torque and speed by its battery and motor size. While it is sometimes desirable to have the ability to print at the fastest speed possible to satisfy customer requirements, the printer's maximum speed (for marketing and other purposes) can be limited by the available motor torque when peeling.

(142) Drive tables can be generated and stored in the printer's memory and accessed when the peeler bar is and/or is not being used. The drive tables can include, for example, drive motor parameters and/or other types of settings that are optimized for both peel mode and non-peel mode. This optimization can be used to, for example, print at a faster speed when in non-peel mode without negatively impacting the print quality and/or other variables while in peel mode. As another example, the drive tables can include data related to the print registration (such as, e.g., where the print line should begin) and/or other data that may be affected based upon whether a peeler bar is engaged or disengaged. Additionally, the manufacture may be able to advertise the maximum print speed at a higher rate than that used when in peel mode. Adjusting the print speed, torque and/or other drive motor parameter can also solve historical problems related to the printer stalling and overheating when peeling. Optimal settings could also be dependent on the particular type of media because it could be dependent on media thickness, shape, material, temperature, adhesive characteristics, label gap indication (including blackmark or notch), and even perforations or other die-cut attributes.

(143) Process **1300** begins at **1302** and advances to **1304** at which the printer receives a command to print/encode indicia onto media. At **1306** the printer determines whether or not a peeler bar is engaged. For example, the printer can include a peeler sensor that generates one or more signals when the peeler bar is engaged (properly or otherwise) and/or when the peeler bar is disengaged (properly or otherwise). The peeler sensor may be configured to generate the signal(s) in response to making physical contact, electrical contact, and/or magnetic contact with locking protrusions **808** and/or in response to detecting the absence of locking protrusions **808**. Information as to whether the peeler is engaged could also be recalled from a memory location where the information was established from some prior process. As another example, regardless of whether a peeler sensor is included and/or functioning properly, the printer may be able to read information (visually, wirelessly, mechanically and/or by any other means) from the media and/or a spindle onto which the media is wound, and determine at **1306** whether or not the peeler bar should be and therefore is engaged.

(144) In response to determining at **1306** the peeler bar is engaged, process **1300** proceeds to **1308** at which the printer can access, for example, a torque setting stored in memory. The torque setting can be used by the printer to optimize print speed for printing/encoding indicia onto media that is

being removed from a backing using the printer's peeler bar. The addition of a detection device, such as a switch to detect when a printer is in peel mode combined with software, firmware and/or other hardware optimized to drive the motor according to the switch status, can permit the maximum speed of a printer to be increased when not peeling and slowed down for additional torque when peeling.

(145) At **1310**, the printer can utilize other printing settings for printing indicia onto media to be peeled from its backing. For example, an optical sensor (e.g., transmissive sensor, reflective sensor, or combination thereof), a proximity sensor, or other type of label-detecting sensor can be used by the printer to determine whether a printed/encoded label is affixed to the peeler bar and awaiting removal.

(146) At **1312**, the printer prints/encodes indicia onto the media in accordance with the one or more printing settings (including, e.g., the torque setting). At **1314**, the peeler bar removes the media from its backing as the printer's roller bars (and/or other components) advance the media out of the printer's printing/encoding zones. At **1316** the printer determines whether it should pause printing. Printing may be paused in response to, for example, receiving a pause command from a user and/or another device, a printer sensor detecting an error has occurred, a printer sensor detecting that the media is affixed to the peeler bar and is awaiting removal, and/or for any other reason.

(147) In response to determining at **1316** that printing should not be paused (e.g., no error detected, nothing blocking the path of the media being printed/encoded, etc.), process **1300** returns to **1312** and the printing continues. In response to determining at **1316** that printing should be paused until, for example, the previously printed media is removed from the peeler bar, the printer waits at **1318** for the printed media to be removed from the peeler bar. A determination is made at **1320** whether the printed/encoded media has been removed from the peeler bar. In response to determining at **1320** the printed/encoded media has not been removed from the peeler bar, process **1300** returns to **1318** and continues to wait. In response to determining at **1320** that the media has been removed from the peeler bar, process **1300** returns to **1304**.

(148) After determining at **1306** that the peeler bar is disengaged, process **1300** advances to **1322** at which the printer utilizes a torque setting for printing onto backless media that is not to be peeled. At **1324**, other print settings can also be used for printing onto backless media and at **1326** the printer can print/encode indicia onto the media in accordance with the printing settings. Process **1300** may then return to **1304**.

(149) In summary, the peeler can be activated and placed into the peeling position by the user and/or the printer when peeling is desired, and then retracted out of the way into a non-peeling position when peeling is complete and/or no longer desired. Some embodiments of the peel bar may have various advantages over similar mechanisms, such as those based on the peeler assembly's compact size, the push-button actuation, and the peel bar's semi-automatic, threadless operation (traditional liner or backing threading is replaced with media threading, i.e., simply placing the peel bar on top of a media unit). The peeler's pivoting arm(s) and roller(s) of some embodiments can be adapted to force the backing of a label around a stripper bar, cams and springs in the media access cover can provide the peeling force, and an actuating lever can also be included. To operate the peeler, the user can open the media door and press an internal (or external) lever or other type of button, thereby releasing the peeler into the extended, ready position using springs included in the printer. As the user closes the media door, pins (e.g., locking protrusions **808**) on the ends of the peeler can be adapted to engage a spring-loaded cam on each side of the media door, engaging the peeler and pulling the backing tight around the peeler bar. When then user and/or printer is finished peeling labels, the user and/or printer can again open the media door, and push or otherwise return the peel bar back into its stowed, non-peeling position.

(150) Universal Printhead

(151) A printer in accordance with embodiments discussed herein can include a universal printhead, such as that shown in FIGS. **14A-14C**. FIG. **14A** shows a portion of a printer that includes, among

other things, universal printhead **806** and various components of a peeler assembly, such as that discussed in connection with FIGS. **8A-8D**. FIGS. **14B** and **14C** show two opposite faces of printhead **806**. In particular, FIG. **14B** shows an example printing face **1402** and FIG. **14C** shows an example mounting face **1404**.

(152) Printhead **806** is shown as being generally rectangular in shape and defined by first distal end **1406**, second distal end **1408** (located opposite first distal end **1406**), top end **1410** and bottom end **1412** (located opposite top end **1410**). (The terms “bottom” and “top” are referenced herein to avoid overcomplicating the discussion. Any connotation or denotation relating to the meaning of “top” and “bottom,” such as those relating to their relative position in relation to the direction of gravitational pull, are not to necessarily be read into their meaning herein.) Printing face **1402** is generally defined by first distal end **1406**, second distal end **1408**, top end **1410** and bottom end **1412**. Printing face **1402** can also include one or more printing components, such as thermal element **1414**. Thermal element **1414** may be configured to, for example, conduct thermal energy and cause at least a portion of a print media and/or print ribbon to be heated.

(153) FIG. **14B** also shows electrical interface **1416**, which may be configured to receive and/or otherwise couple with a connector component, such as a serial and/or parallel bus. Electrical interface **1416** may enable printhead **806** to receive inputs (such as printing commands) from, for example, control circuitry (e.g., processor **1104**). Although electrical interface **1416** is shown as being positioned proximate to bottom end **1412**, electrical interface **1416** may be positioned in any suitable location(s) that enables it to, e.g., be coupled to thermal element **1414**.

(154) As shown in FIG. **14C**, universal printhead **806** can include one or more mounting components configured to mount printhead **806** to a printer. For example, a first mounting component, such as mounting holes **1420**, and/or a second mounting component, such as one or more grooves **1422**, can be included on mounting face **1404**.

(155) Mounting holes **1420** can be unthreaded or threaded. In some embodiments, there may be one or more of each type of mounting holes. In other embodiments, the mounting holes may be of the same type (e.g., threaded or unthreaded). The size (e.g., diameter and depth) of mounting holes **1420** may be the same, different or a combination thereof (e.g., some the same, some different).

(156) Grooves **1422** are shown as extending substantially parallel to top end **1410** and the bottom end **1412**. In some embodiments, one or more grooves can instead or additionally be positioned in any suitable fashion on mounting face **1404**.

(157) Additionally or alternatively, universal printhead **806** can also include one or more notches, heat sinks, shielding components, and/or any other mounting component(s). As such, mounting face **1404** of universal printhead **806** can provide a generic mounting platform that has flexible mechanical interface to compliment the flexible electrical interface provided by electrical interface **1416**. As such, universal printhead **806** may be configured to be used with different printers (e.g., different manufacturers' printers, types of printers, etc.). Additionally, mounting face **1404** can be configured to facilitate relatively easy replacement of universal printhead **806** over the printer's product life as compared to other printheads. In some embodiments, printhead **806** can be made at least partially from aluminum and/or any other suitable material(s).

## CONCLUSION

(158) Various other features for, modifications to and other embodiments of the inventions set forth herein will come to mind to one skilled in the art to which these inventions pertain having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. For example, while examples discussed herein are often related to mobile printers, one skilled in the art would appreciate that other types of printers, such as desktop or less mobile printers, as well as other types of devices may benefit from embodiments discussed herein. Therefore, it is to be understood that the inventions are not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included herein. Although specific terms

are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

## Claims

1. A method comprising: positioning a peeler assembly of a printer in one of a peeling position, in which media is peeled from a backing, and a non-peeling position, in which the backing and the media remain affixed to each other; detecting a portion of the peeler assembly to determine a current position of the peeler assembly; and adjusting at least one print setting in response to detecting the current position of the peeler assembly.
2. The method of claim 1, wherein detecting the current position is performed by a sensor and adjusting the at least one print setting is performed by a processor.
3. The method of claim 2, wherein the at least one print setting comprises at least one of print speed, printhead temperature, or printhead position.
4. The method of claim 3, wherein the sensor is carried by a media cover of the printer.
5. The method of claim 2, wherein the sensor comprises a proximity sensor.
6. The method of claim 2, wherein the sensor comprises a proximity sensor.
7. The method of claim 1, further comprising: providing an output to a user corresponding to the current position of the peeler assembly.
8. The method of claim 7, wherein the output includes at least one of the list including an audible notification, a visual notification, or a mechanical notification.
9. The method of claim 1, further comprising: storing the current position of the peeler assembly in memory.
10. The method of claim 1, further comprising: presenting the current position of the peeler assembly to a user via a user interface.
11. The method of claim 1, wherein the peeler assembly includes locking protrusion that is configured to selectively mate with a latch of the printer to lock peeler assembly in one of the peeling position of the non-peeling position.
12. The method of claim 11, wherein detecting a portion of the peeler assembly to determine a current position of the peeler assembly comprises: detecting a presence or absence of the locking protrusions when the peeler assembly is in the peeling position or the non-peeling position.
13. A method comprising: positioning a peeler assembly of a printer in one of a peeling position, in which media is peeled from a backing, and a non-peeling position, in which the backing and the media remain affixed to each other; detecting a current position of the peeler assembly; and adjusting at least one print setting in response to detecting the current position of the peeler assembly, wherein detecting the current position is performed by a sensor and adjusting the at least one print setting is performed by a processor, and wherein the at least one print setting comprises at least one of print speed, printhead temperature, or printhead position.
14. The method of claim 13, further comprising: providing an output to a user corresponding to the current position of the peeler assembly.
15. The method of claim 14, wherein the output includes at least one of the list including an audible notification, a visual notification, or a mechanical notification.
16. The method of claim 15, wherein the sensor is carried by a media cover of the printer.
17. The method of claim 14, wherein the sensor comprises a proximity sensor.
18. The method of claim 13, further comprising: storing the current position of the peeler assembly in memory.
19. The method of claim 13, further comprising: presenting the current position of the peeler assembly to a user via a user interface.
20. A method comprising: positioning a peeler assembly of a printer in one of a peeling position, in which media is peeled from a backing, and a non-peeling position, in which the backing and the

media remain affixed to each other; detecting a current position of the peeler assembly; adjusting at least one print setting in response to detecting the current position of the peeler assembly; and providing an output to a user corresponding to the current position of the peeler assembly.

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